JEE (MAIN) : ENTHUSIAST COURSE SCORE -II

Time : 3.00 Hours **Maximum Marks : 300**

Q2

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet and OMR sheet (Side-1 & Side-2) with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of 90 questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics**, **Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
   (a) Section-I contains 20 **multiple choice** questions with **only one correct** option.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
   (b) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray marks on the Answer Sheet.**
9. **Take g = 10 m/s^2 unless otherwise stated.**

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

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Topic : Full Syllabus

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. The temperature of 5 mole gas changed from 100°C to 120°C. Change in internal energy is 80 J. The total heat capacity of the gas at constant volume will be :-

(A) $8\ JK^{-1}$ (B) $0.8\ JK^{-1}$

(C) $4\ JK^{-1}$ (D) $0.4\ JK^{-1}$

2. 540 g of ice at 0°C is mixed with 540 g water at 80°C. The final temperature of the mixture is :-

(A) 0°C (B) 40°C

(C) 80°C (D) less than 0°C

3. Two identical thin rings, each of radius R meter are coaxially placed at distance R meter apart. If Q_1 and Q_2 coulomb are respectively the charges uniformly spread on the two rings, the work done in moving a charge q from the centre of one ring to that of the other is :-

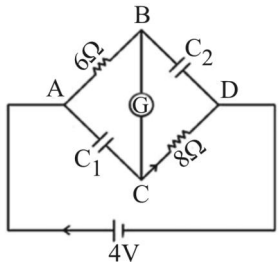
(A) Zero

(B) $q(Q_2 - Q_1)(\sqrt{2}-1)/(\sqrt{2}.4\pi\varepsilon_0 R)$

(C) $q\sqrt{2}(Q_1 + Q_2)/4\pi\varepsilon_0 R$

(D) $q(Q_1 - Q_2)(\sqrt{2}+1)/(\sqrt{2}.4\pi\varepsilon_0 R)$

4. In this figure the resistance of the coil of galvanometer G is 2Ω. The emf of the cell is 4V. The ratio of potential difference across C_1 and C_2 is:



(A) 1 (B) $\frac{4}{5}$ (C) $\frac{3}{4}$ (D) $\frac{5}{4}$

5. A copper wire of length ' ℓ ' and radius 'r' is nickel plated till its final radius is 2r. If the resistivity of the copper and nickle are ρ_c and ρ_n, then find the equivalent resistance of wire between its ends ?

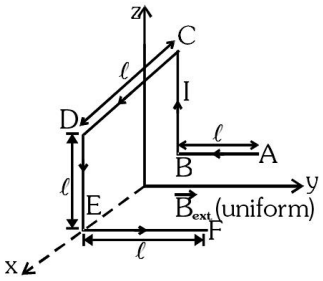
(A) $\frac{\rho_c \rho_n \ell}{\pi r^2(\rho_n + 3\rho_c)}$ (B) $\frac{\rho_c \rho_n \ell}{\pi r^2(\rho_n + 4\rho_c)}$

(C) $\frac{\rho_c \rho_n \ell}{4\pi r^2(\rho_n + 3\rho_c)}$ (D) None of these

6. A solenoid of 10 henry inductance and 2 ohm resistance, is connected to a 10 volt battery. In how much time the magnetic energy will increases to one forth of the maximum value :–

(A) 3.5 s (B) 2.5 s (C) 5.5 s (D) 7.5 s

7. A wire ABCDEF with each side of length ' ℓ ' bent as shown in figure and carrying a current I. If it is placed in a uniform magnetic field B which is parallel to +y direction. Magnetic force experienced by the wire is:-



(A) $BI\ell$, along + z direction

(B) $BI\ell$, along $-$z direction

(C) $2BI\ell$, along + z direction

(D) $2BI\ell$, along $-$z direction

8. An L–C–R circuit has L = 10 mH. R = 3 ohm and C = 1μF connected in series to a source of V = 15 cos ωt volt. The current-amplitude and the average power dissipated per cycle at a frequency 10% lower than the resonant frequency will respectively be :-

(A) 0.704A, 0.744W (B) 0.704A, 0.704W

(C) 7.04A, 7.44W (D) 70.4A, 74.4W

9. A radio wave of frequency 90 MHz (FM) enters a ferrite rod. In ferrite, $\varepsilon_r = 10^3$ and $\mu_r = 10$, then the velocity and wavelength in ferrite are :-

(A) 3×10^6 ms^{-1}, 3.33×10^{-2} m

(B) 3×10^6 ms^{-1}, 3.33×10^{-4} m

(C) 3×10^6 ms^{-1}, 3.33×10^{-3} m

(D) None of the above

10. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A - 1.50)$ eV and de-Broglie wavelength is λ_B. If the de-Broglie wavelengths of these photoelectrons are related as $\lambda_B = 2\lambda_A$, then :-

(a) The work function of A is 2.25 eV

(b) The work function of B is 4.20 eV

(c) $T_A = 2.00$ eV

(d) $T_B = 2.75$ eV

(A) a, c (B) b, d

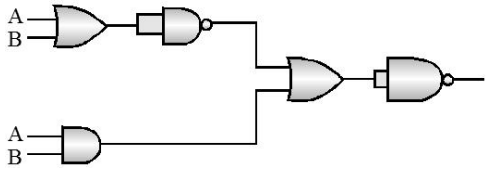
(C) a, b, c (D) b, c, d

11. With what potential an electron should be accelerated so that its de Broglie wave-length becomes equal to the wave length of second line of Balmer series for He^+ ion ?

(A) $\frac{9R^2h^2}{32me}$ (B) $\frac{162R^2h^2}{25me}$

(C) $\frac{81R^2h^2}{32me}$ (D) $\frac{32R^2h^2}{81me}$

12. Output for the following Boolean circuit is :



(A) $\overline{AB}$ (B) $A\bar{B} + \bar{A}B$

(C) $\bar{A}\bar{B} + AB$ (D) $A + B + \overline{AB}$

13. Two beams of light having intensities I and 4I interfere to produce a fringe pattern on a screen. The phase difference between the beam is $\frac{\pi}{2}$ at point A and 2π at point B. Then find out the difference between the resultant intensities at A and B.

(A) 2I (B) 5I (C) I (D) 4I

14. If the length of rod A is 3.25 ± 0.01 cm and that of B is 4.19 ±0.01 cm, then the rod B is longer than rod A by :-

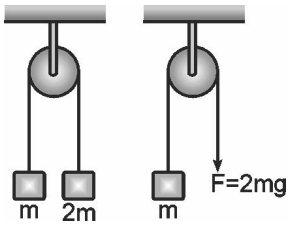
(A) 0.94 ± 0.00 cm

(B) 0.94 ± 0.01 cm

(C) 0.94 ± 0.02 cm

(D) 0.94 ± 0.005 cm

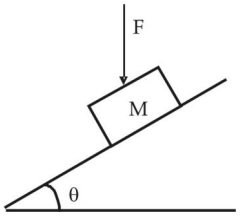
15. Figure shows two pulley arrangements for lifting a mass m. In (a) the mass is lifted by attaching a mass 2 m while in (b) the mass is lifted by pulling the other end with a downward force F = 2 mg, If f_a & f_b are the accelerations of the two masses then (Assume string is massless and pulley is ideal)



(A) $f_a = f_b$ (B) $f_a = f_b/2$

(C) $f_a = f_b/3$ (D) $f_a = 2\ f_b$

16. A block of mass M is sliding down the plane. Coefficient of static friction is μ_s and kinetic friction is μ_k. Then friction force acting on the block is :-



(A) $\mu_s Mg \cos\theta$

(B) $(F + Mg)\sin\theta$

(C) $\mu_k (F + Mg)\cos\theta$

(D) $(Mg + F)\tan\theta$

17. A particle travels such that $\frac{dV}{dt} = -a\sqrt{v}$ where a is constant. Its initial velocity is v_0. The time at which it stops is :-

(A) $\frac{2\sqrt{V_0}}{a}$ (B) $\frac{\sqrt{V_0}}{a}$

(C) $2a\sqrt{V_0}$ (D) $a\sqrt{V_0}$

18. The centre of mass of a non uniform rod of length L whose mass per unit length $\lambda = \left(\frac{Kx^2}{L}\right)$ where K is a constant and x in the distance from one end is :-

(A) $\frac{3L}{4}$ (B) $\frac{L}{8}$ (C) $\frac{L}{K}$ (D) $\frac{3L}{K}$

19. A circular disc of radius R and thickness $\frac{R}{6}$ has moment of inertia I about an axis passing through its centre and perpendicular to its plane. It is melted and recasted into a solid sphere. The moment of inertia of the sphere about its diameter as axis of rotation is:-

(A) I (B) $\frac{2I}{8}$ (C) $\frac{I}{5}$ (D) $\frac{I}{10}$

20. A satellite is orbiting the earth in a circular orbit of radius r. Its :-

(a) Kinetic energy varies as $\frac{1}{r}$

(b) Angular momentum varies as $\frac{1}{\sqrt{r}}$

(c) Linear momentum varies as $\frac{1}{\sqrt{r}}$

(d) Frequency of revolution varies as $\frac{1}{r^{3/2}}$

(A) Only d (B) a, c, d

(C) Only a, b (D) Only b

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

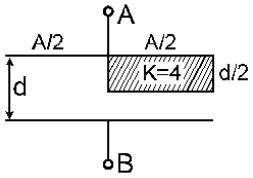
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1. If the volume of a block of metal changes by 0.12% when it is heated through 20°C, the coefficient of linear expansion (in $°C^{-1}$) of the metal is $x \times 10^{-5}$, find the value of x ?

2. The volume of a metal sphere increases by 0.24% when its temperature is raised by 40°C. The coefficient of linear expansion of the metal is $x \times 10^{-5}/°C$, find the value of x ?

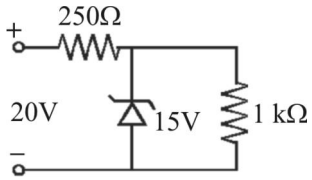
3. The electric field in a region of space is given by $E = (5\hat{i} + 2\hat{j})$ N/C. The electric flux through an area of $2m^2$ lying in the YZ plane, in S.I. units is :

4. The equivalent capacitance between terminals 'A' and 'B' is $\frac{\alpha}{\beta}\frac{\in_0 A}{d}$, α & β being least integer. Determine the value of $\alpha + \beta$.
(letters have their usual meaning)



5. Mass defect in a $_{11}Na^{23}$ atom is 0.21 amu. Then the binding energy per nucleon 8.505×10^{x} eV than 'x' is :

6. A zener diode, having breakdown voltage equal to 15 V, is used in a voltage regulator circuit shown in figure. The current through the zener diode is (in mA) :

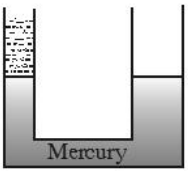


7. Light travel in two medium M_1 & M_2 with speed 1.5×10^8 m/s and 2×10^8 m/s respectively.
The critical angle between them is $\tan^{-1}\left(\frac{x}{\sqrt{7}}\right)$.
Find x.

8. A tuning fork gives 4 beats with 50cm length of a sonometer wire. If the length of the wire is shortened by 1cm, the number of beats is still the same. The frequency of the fork (in Hz) is :-

9. Two simple pendulums of length 5 m and 20 m respectively are given small linear displacement in one direction at the same time. They will again be in phase when the pendulum of shorter length has completed..... oscillations.

10. A U-tube in which the cross-sectional area of the limb on the left is one quarter, the limb on the right contains mercury (density 13.6 g/cm^3). The level of mercury in the narrow limb is at a distance of 36 cm from the upper end of the tube. What will be the rise in the level of mercury in the right limb if the left limb is filled to the top with water (in cm)



Topic : Full Syllabus

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. At constant temperature, the equilibrium constant (K_P) for the decomposition reaction $N_2O_4 \rightleftharpoons 2NO_2$ is expressed by $K_P=4x^2P/(1-x^2)$ where P is pressure, x is extent of decomposition. Which of the following statement is true ?

(A) K_P increases with increase of P

(B) K_P increases with increase of x

(C) K_P increases with decrease of x

(D) K_P remains constant with change in P or x

2. Which of the following is true statement ?

(A) Rate of reaction does not changes on changing stoichiometry of reaction

(B) Rate can be negative

(C) If rate of appearance of NH_3 is 1.7 kg/hr in the reaction $N_2 + H_2 \longrightarrow NH_3$ then rate of disappearance of H_2 will be 42.5 gm/min

(D) Rate of disappearance of reactants or rate of appearance of products does not changes on changing stoichiometry of the reaction

3. In 500 ml of 0.2M-NaCl solution, 1500 ml of 0.4M $MgCl_2$ solution is mixed. The only incorrect final concentration of ion is (assume volume to be additive) :

(A) $[Na^+] = 0.2$ M (B) $[Mg^{2+}] = 0.3$ M

(C) $[Cl^-] = 0.65$ M (D) $[Mg^{2+}] = 7.2$ g/L

4. The solubility of PbF_2 (formula weight = 245) is 0.46g/L. What is the solubility product :–

(A) 1.1×10^{-10} (B) 2.6×10^{-8}

(C) 1.1×10^{-7} (D) 6.8×10^{-9}

5. The vapour pressure of CS_2 at 50°C is 854 torr and a solution of 2.0 gm sulphur in 100 gm of CS_2 has vapour pressure 848.9 torr. If the formula of sulphur molecule is S_n, then calculate the value of "n" (approx) (At. mass of S = 32) :- [Assume very dilute solution]

(A) 2 (B) 4

(C) 8 (D) 6

6. 18 gm glucose is completely combusted in bomb calorimeter, of heat capacity 1400 kJ/K, temperature changes from 27°C to 27.2 °C magnitude of standard enthalpy of combustion of glucose in kJ/mol. [R = 8.314 J/mol-K]

(A) 1200 (B) 2000

(C) 2800 (D) 1400

7. To which of the following atom, the attachment of electron is most difficult ?

(A) Radon (B) Nitrogen

(C) Oxygen (D) Radium

8. Which of the following statement is not correct ?

(A) Lu^{+3} has the strongest tendency toward complex formation among trivalent lanthanoid ions

(B) Ce has maximum composition in misch metal

(C) f-block elements can have electrons from f^0 to f^{14}

(D) Nd, Np and Nb all are f-block elements

9. Why is HCl not used to make the medium acidic in oxidation reactions of $KMnO_4$ in acidic medium?

(A) Both HCl and $KMnO_4$ act as oxidising agents.

(B) $KMnO_4$ oxidises HCl into Cl_2 which is also an oxidising agent.

(C) $KMnO_4$ is a weaker oxidising agent than HCl.

(D) $KMnO_4$ acts as a reducing agent in the presence of HCl.

10. The correct order of the O–O bond length in O_2, H_2O_2 and O_3 is :-

(A) $O_2 > O_3 > H_2O_2$ (B) $O_3 > H_2O_2 > O_2$

(C) $H_2O_2 > O_3 > O_2$ (D) $H_2O_2 > O_2 > O_3$

11. Which of the following complex have t_{2g}^4, e_g^2 configuration ?

(A) $[CoF_6]^{-3}$ (B) $[Co(H_2O)_6]^{+3}$

(C) $[Co(C_2O_4)_3]^{-3}$ (D) $[Co(CN)_6]^{-3}$

12. Which of the following complex will show optical activity ?

(A) trans-$[Co(NH_3)_4Cl_2]^+$

(B) $[Cr(H_2O)_6]^{3+}$

(C) Cis-$[Co(NH_3)_2(en)_2]^{3+}$

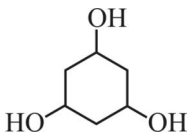
(D) trans-$[Co(NH_3)_2\ (en)_2]^{3+}$

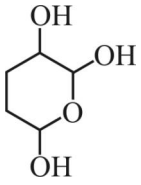
13. The orbital angular momentum of electron in 'd' orbital is equal to :-

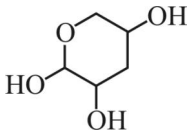
(A) $\sqrt{2}\hbar$ (B) $2\sqrt{3}\hbar$

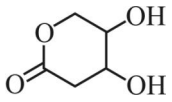
(C) 0 h (D) $\sqrt{6}\hbar$

14. In which molecule, cleavage by HIO_4 is not observed ?

(A) 

(B) 

(C) 

(D) 

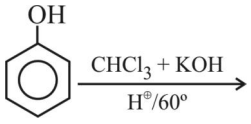
15. Match the following reactants in Column I with the corresponding reaction products in Column II :-

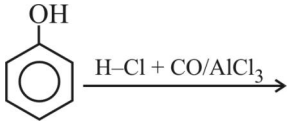
Column-I		Column-II	
(a)	Benzene + Cl_2 $\xrightarrow{AlCl_3}$	(1)	Benzoic acid
(b)	Benzene + CH_3Cl $\xrightarrow{AlCl_3}$	(2)	Methyl Phenyl ketone
(c)	Benzene + CH_3COCl $\xrightarrow{AlCl_3}$	(3)	Toluene
(d)	Toluene $\xrightarrow{KMnO_4 / NaOH}$	(4)	Chlorobenzene
		(5)	Benzene hexachloride

Codes

	a	b	c	d
(A)	4	3	2	1
(B)	1	2	3	4
(C)	4	3	1	5
(D)	1	5	2	3

16. Which of the following is not correctly matched :-

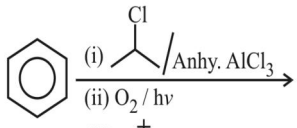
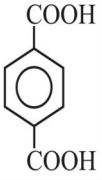
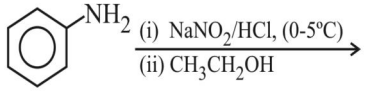
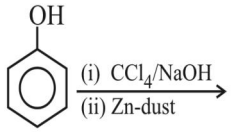
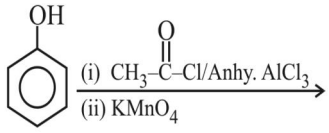
(A)  Reimer-Tiemann formylation

(B)  Gatterman Koch reaction

(C) $R-N\equiv C \xrightarrow{CHCl_3 + KOH}$ Isocyanide test

(D) $R-N\equiv C \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2 + HCl}$ Mendius Reaction

17.

	List-I		List-II
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

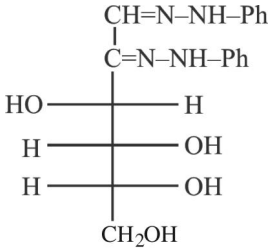
(A) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4

(B) P ⟶ 4;Q ⟶ 3;R ⟶ 2;S ⟶ 1

(C) P ⟶ 2;Q ⟶ 4;R ⟶ 3;S ⟶ 1

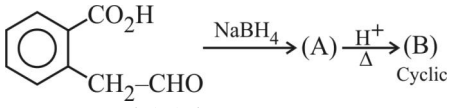
(D) P ⟶ 2;Q ⟶ 3;R ⟶ 4;S ⟶ 1

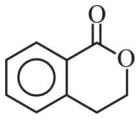
18. The given osazone can be obtained by :-

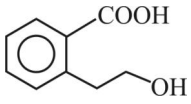


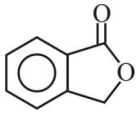
(A) D–glucose (B) D–mannose

(C) D–Idose (D) Both (A) & (B)

19. 
Compound (B) is :-

(A) 

(B) 

(C) 

(D) (A) and (C) both

20. $CH_3CH{=}CHCHO \xrightarrow[\text{Aldol condensation}]{OH^-} \xrightarrow{\Delta}$;
Product A is :-

(A) $CH_3(CH = CH)_3CHO$

(B) $CH_3CH_2CH_2(CH{=}CH)_2CHO$

(C) $CH_3(CH_2CH_2)_3CH{=}CH{-}CHO$

(D) None is correct

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

1. Calculate value of λ^0_m ($\Omega^{-1}cm^2mole^{-1}$) for $CaCl_2$ from following data :

Conc. (mole/litre)	0.25	1
λ_m ($\Omega^{-1}cm^2mole^{-1}$)	300	280

2. 3 moles of $FeSO_4$ are oxidised by "a" mole of $K_2Cr_2O_7$ in acidic medium whereas 3 moles of FeC_2O_4 are oxidised by "b" mole of $K_2Cr_2O_7$ in acidic medium, the ratio of "b" and "a" is :-

3. A quantity of aluminium has a mass of 54 gm. What is the mass of same number of magnesium atoms?
(Given : At. wt. of Mg = 24)

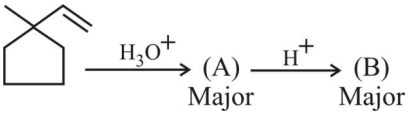
4. For which orbit number of Be^{3+}, value of radius is equal to 0.529 Å ?

5. Count basic hydroxides NaOH, ClOH, KOH, $Ca(OH)_2$ $Zn(OH)_2$, $(OH)_2SO_2$, $HO(NO_2)$, $Al(OH)_3$

6. Find the total number having both p_π-p_π & p_π-d_π type of bond.

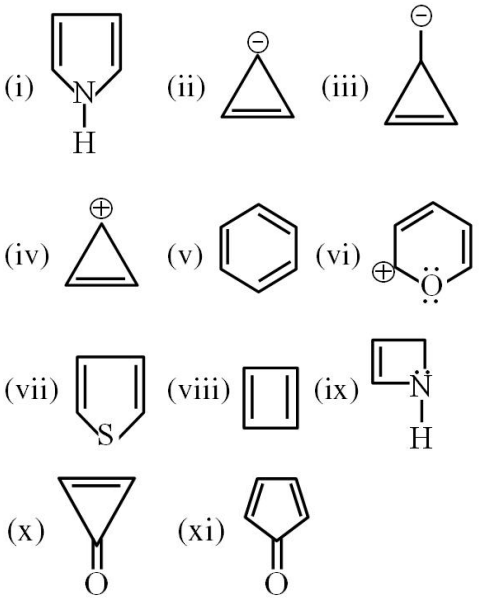
SO_3, SO_2, SO_4^{2-}, SO_3^{2-}, XeO_3, XeO_4, $POCl_3$, SNF_3

7. In the following reaction

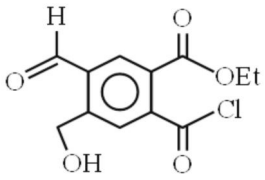


The number of α-hydrogen present in B compound.

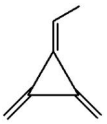
8. How many of the following compounds is are aromatic ?



9. How many moles of RMgX consumed in the given compound ?



10.



Total number of geometrical isomers of above compound are

Topic : Full Syllabus

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

---

1. If $a + b + c = 3$ and $a > 0, b > 0, c > 0$ then the greatest value of $a^2b^2c^2$ is :-

(A) $\frac{3^{10}2^4}{7^7}$ (B) 1

(C) $\frac{3^8 2^4}{7^7}$ (D) None of these

2. If α, β are roots of eq. $375x^2 - 25x - 2 = 0$ & $S_n = \alpha^n + \beta^n$ then the value of $\lim_{n\to\infty} \sum_{r=1}^{n} S_r =$

(A) $\frac{3}{4}$ (B) $\frac{1}{12}$ (C) 0 (D) 15

3. If A is matrix of order (3×3) and $|A| = 2$ then value of $|\text{adj adj adj adj adj } (2A)|$ is equal to : –

(A) 2^9 (B) 2^{128}

(C) 2^{32} (D) None

4. Let 13 seats in a straight line are to be occupied by 4 persons. The number of ways in which there should be exactly four empty seats and no other person between two particular persons is-

(A) $15(^6C_3)$ (B) $30(^6C_3)$

(C) $16(^7C_2)$ (D) $32(^7C_2)$

5. In a pack of 52 cards one by one cards are randomely drawn untill all red court cards are not drawn, then find the probability that this process is completed in 17th draw

(A) $\frac{^{12}C_6.^{40}C_{11}}{^{52}C_{16}.^{36}C_{17}}$ (B) $\frac{1}{46}\frac{^6C_5.^{46}C_{11}}{^{52}C_{16}}$

(C) $\frac{^{46}C_{11}}{^{52}C_{16}.6}$ (D) None

6. The value of $i^{1+3+5+...+(2n+1)}$ is :-

(A) i if n is even, –i if n is odd

(B) 1 if n is even, –1 if n is odd

(C) 1 if n is odd, –1 if n is even

(D) i if n is even, 1 if n is odd

7. If the maximum value of $y = \frac{7 + 6\tan x - \tan^2 x}{(1 + \tan^2 x)}$ is λ then the value of $\log_{\sqrt{2}}(\lambda)$ is :

(A) 0 (B) 6 (C) 8 (D) 1

8. S.D. of $x_1, x_2, ...x_n$ is 2, if $\sum_{i=1}^{n} x_i = 20$ and $\sum_{i=1}^{n} x_i^2 = 100$. Then value of n is

(A) 5 or 15 (B) 10 or 20

(C) 5 or 10 (D) 5 or 20

9. Let R be the relation on the set R of all real numbers defined by a R b iff $|a-b| \leq 1$. Then R is :-

(A) Reflexive and symmetric

(B) Symmetric only

(C) Transitive only

(D) Anti symmetric only

10. Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$. If [a, b] is the range of the function then $4a - b$ is equal to:

(A) 11 (B) $11 - \pi$

(C) $11 + \pi$ (D) $15 - \pi$

11. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right) + \cos^{-1}\left(\cos\frac{7\pi}{6}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to :

(A) $\frac{11\pi}{12}$ (B) $\frac{17\pi}{12}$

(C) $\frac{31\pi}{12}$ (D) $-\frac{3\pi}{4}$

12. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 7 & , x \leq 1 \\ -2x + \log_2(b^2 - 4) & , x > 1 \end{cases}$

Then the set of all values of b, for which f(x) has maximum value at x = 1, is :

(A) $(-6, -2)$

(B) $(2, 6)$

(C) $[-6, -2) \cup (2, 6]$

(D) $[-\sqrt{6}, -2) \cup (2, \sqrt{6}]$

13. Let λ^* be the largest value of λ for which the function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is increasing for all $x \in R$. Then $f_\lambda{}^*(1) + f_\lambda{}^*(-1)$ is equal to :

(A) 36 (B) 48

(C) 64 (D) 72

14. The area of the region

$S = \{(x, y) : y^2 \leqslant 8x, y \geqslant \sqrt{2}x, x \geqslant 1\}$ is :

(A) $\frac{13\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$

(C) $\frac{5\sqrt{2}}{6}$ (D) $\frac{19\sqrt{2}}{6}$

15. Let the solution curve of the differential equation $xdy = \left(\sqrt{x^2 + y^2} + y\right) dx$, $x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

16. For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$, if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

(A) $3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

(B) $3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

(C) $3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

(D) $3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

17. Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$, $\alpha, \beta \in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $|\vec{b}|^2 = 6$, If $\vec{a} \cdot \vec{b} = 3\sqrt{2}$, then the value of $(\alpha^2 + \beta^2)\,|\vec{a} \times \vec{b}|^2$ is equal to

(A) 90 (B) 75

(C) 95 (D) 85

18. For $0 < q < \pi/2$, if the eccentricity of the hyperbola $x^2 - y^2\text{cosec}^2\theta = 5$ is $\sqrt{7}$ times eccentricity of the ellipse $x^2\,\text{cosec}^2\theta + y^2 = 5$, then the value of θ is :

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{12}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

19. Let C : $x^2 + y^2 = 4$ and C' : $x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C' intersect at two distinct points, is **R**– [a, b], then the point $(8a + 12, 16b - 20)$ lies on the curve :

(A) $x^2 + 2y^2 - 5x + 6y = 3$

(B) $5x^2 - y = -11$

(C) $x^2 - 4y^2 = 7$

(D) $6x^2 + y^2 = 42$

20. If the shortest distance of the parabola $y^2 = 4x$ from the centre of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$ is d, then d^2 is equal to :

(A) 16 (B) 24

(C) 20 (D) 36

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

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1. The number of distinct real solution of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

in the interval $\frac{-\pi}{4} \leqslant x \leqslant \frac{\pi}{4}$ is-

2. The number of rational term in expansion of $\left(5^{\frac{1}{2}} + 7^{\frac{1}{6}}\right)^{65}$ is :-

3. The value of

$$\frac{\left(1+\sqrt{3}\tan 1^\circ\right)\left(1+\sqrt{3}\tan 2^\circ\right)(\tan 1^\circ + \tan 59^\circ)(\tan 2^\circ + \tan 58^\circ)}{(1+\tan^2 1^\circ)(1+\tan^2 2^\circ)}$$

is :-

4. The sum of the maximum and minimum values of the function $f(x) = |5x - 7| + [x^2 + 2x]$ is the interval $\left[\frac{5}{4}, 2\right]$, where [t] is the greatest integer $\leq t$ is _____

5. Let $f(x) = |(x-1)(x^2-2x-3)| + x - 3$, $x \in R$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval (0, 4), then m + M is equal to _____.

6. The value of the integral

$$\frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi x^2}{2}-x^3\right)\frac{\sin x}{1+\cos^2 x}dx$$

is equal to _____.

7. If

$$n(2n+1)\int_0^1 (1-x^n)^{2n}dx$$

$$= 1177\int_0^1 (1-x^n)^{2n+1}dx,$$

then $n \in N$ is equal to ____

8. Let

$$\underset{0\le x\le 2}{\text{Max}}\left\{\frac{9-x^2}{5-x}\right\} = \alpha \text{ and } \underset{0\le x\le 2}{\text{Min}}\left\{\frac{9-x^2}{5-x}\right\} = \beta$$

If

$$\int_{\beta-\frac{8}{3}}^{2\alpha-1} \text{Max}\left\{\frac{9-x^2}{5-x}, x\right\}dx = \alpha_1 + \alpha_2 \log_e\left(\frac{8}{15}\right)$$

then $\alpha_1 + \alpha_2$ is equal to _____

9. A line passes through A(4, −6, −2) and B(16, −2, 4). The point P(a, b, c) where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points P(a, b, c) and Q(4, −12, 3) is equal to ____.

10. Let A(−2, −1), B(1, 0), C(α, β) and D(γ, δ) be the vertices of a parallelogram ABCD. If the point C lies on 2x − y = 5 and the point D lies on 3x − 2y = 6, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

SPACE FOR ROUGH WORK

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Main)
SCORE - II
14-03-2024

JEE (MAIN) : ENTHUSIAST COURSE SCORE -II

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	B	A	A	A	A	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	D	C	C	C	A	A	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2	2	10	23	6	5	3	396	2	56

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	A	B	C	C	A	D	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	D	A	A	D	C	D	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	320	3	48	2	3	2	10	6	6	0

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	D	C	D	B	D	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	D	B	B	A	A	C	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1	0	3	15	3	6	24	34	22	32

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

$dU = mCvdT$

$\therefore mCv = \frac{dU}{dT} = \frac{80}{20} = 4\ J/K$

4. Ans (B)

At steady state, current in the circuit is

$i = \frac{4V}{6+2+8} = \frac{1}{4}A$

Voltage across C_1 is

$V_1 = V_{AC} = i(6\Omega + 2\Omega) = \frac{1}{4} \times 8 = 2V$

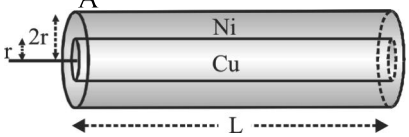
Voltage across C_2 is

$V_2 = V_{BD} = i(2\Omega + 8\Omega) = \frac{1}{4} \times 10 = 2.5V$

$\Rightarrow \frac{V_1}{V_2} = \frac{2}{2.5} = \frac{4}{5}$

5. Ans (A)

$R = \rho\frac{\ell}{A}$



Resistance of copper wire $R_{Cu} = \rho_c \frac{\ell}{\pi r^2}$

$(\because A = \pi r^2)$

Resistance of Nickle wire

$\because A_{Ni} = \pi(2r)^2 - \pi r^2 = 3\pi r^2$

$\Rightarrow R_{Ni} = \rho_n \frac{\ell}{3\pi r^2}$

Both wire are connected in parallel. So equivalent resistance

$R = \frac{R_{Cu}R_{ni}}{R_{Cu}+R_{ni}} = \left(\frac{\rho_C\rho_n}{3\rho_C + \rho_n}\right)\frac{\ell}{\pi r^2}$

6. Ans (A)

$\frac{1}{2}Li^2 = \frac{1}{4}\left(\frac{1}{2}Li_0^2\right)$

$i = \frac{i_0}{2}$

$i = i_0(1 - e^{-t/\tau})$

7. Ans (A)

$\vec{F} = i\left(\vec{\ell} \times \vec{B}\right) = I\left(\ell\hat{i} \times B\hat{j}\right)$

$= IB\ell\hat{k}$

8. Ans (A)

Resonant frequency, $\omega_R = \frac{1}{\sqrt{(LC)}}$

$= \frac{1}{\sqrt{\left(10 \times 10^{-3}H\right) \times \left(1 \times 10^{-6}F\right)}}$

$= 10^4$ per sec

The frequency 10% lower than this is

$\omega = 10^4 - 10^4 \times \frac{10}{100} = 9 \times 10^3$ per sec.

At this frequency, we have

$X_L = \omega L = 9 \times 10^3 \times (10 \times 10^{-3}) = 90$ ohm

$X_C = \frac{1}{\omega c} = 111.11\Omega,$

$Z = \sqrt{R^2 + (X_L - X_C)^2} = 21.32W$

$i_0 = \frac{V_0}{Z} = 0.704A,\ \cos\phi = \frac{R}{Z} = 0.141$

$P_{av} = \frac{V_0 i_0}{2}\cos\phi$

$= \frac{1}{2} \times 15 \times 0.704 \times 0.141 = 0.744W$

10. Ans (C)

$T_A = 4.25 - \phi_A$ ...(1)

$T_B = 4.70 - \phi_B$ ...(2)

$T_B = T_A - 1.5$ ...(3)

and

$\frac{\lambda_A}{\lambda_B} = \frac{1}{2}$ ...(4)

$\because \lambda \alpha \frac{1}{\sqrt{K}} \Rightarrow \frac{K_A}{K_B} = \left(\frac{\lambda_B}{\lambda_A}\right)^2 = \frac{4}{1}$

$\Rightarrow T_A = 4T_B$

from eq. (3) we have

$T_A = 2.0$ eV and $T_B = 0.5$ eV

from eq. (1), $\phi_A = 2.25$ eV

and from eq. (2) $\phi_B = 4.2$ eV

11. Ans (A)

$\frac{1}{\lambda} = R(4)\left[\frac{1}{4} - \frac{1}{16}\right]$

$\frac{1}{\lambda} = R(4) \times \frac{3}{16}$

$\lambda = \frac{4}{3R} = \frac{5}{\sqrt{2meV}}$

$\frac{16}{9R^2} = \frac{h^2}{2meV}$

$v = \frac{9R^2h^2}{32me}$

12. Ans (B)

$\overline{\overline{A + B} + A.B}$

$= \overline{\overline{A + B}}.\overline{A.B}$

$= (A + B).(\bar{A} + \bar{B})$

$= A.\bar{A} + A.\bar{B} + B.\bar{A} + B.\bar{B}$

$A.\bar{B} + B.\bar{A}$

13. Ans (D)

$I_A = I + 4I + 2\sqrt{I.4I}.\cos\frac{\pi}{2} = 5I$

$I_B = I + 4I + 2\sqrt{I.4I}.\cos 2\pi = 9I$

$I_B - I_A = 9I - 5I = 4I$

15. Ans (C)

$f_a = \left(\frac{2m - m}{2m + m}\right) g = \frac{g}{3}$

$f_b = \frac{2mg - mg}{m} = g$

So, $f_a = f_b/3$.

16. Ans (C)

$N = (F + Mg)\cos\theta$

$\therefore$ Friction force $= \mu_k (F + mg)\cos\theta$

17. Ans (A)

$\frac{dV}{dt} = -a\sqrt{v}$

$\frac{dV}{v^{1/2}} = -adt$

$\int v^{-1/2}dv = \int -adt$

$2\sqrt{v} = -at + c$

at $t = 0$ $v = V0$

$2\sqrt{V_0} = c$

$\therefore 2\sqrt{v} = -at + 2\sqrt{V_0}$

put $v = 0$

$t = \frac{2\sqrt{V_0}}{a}$

18. **Ans (A)**

$$x\ cm = \frac{\int x.\,dm}{\int dm}$$

Mass per unit length $\frac{dm}{dx} = \lambda = \frac{Kx^2}{L}$

$$dm = \frac{Kx^2 dx}{L}$$

$$x_{cm} = \frac{\int_0^L x\left(\frac{Kx^2}{L}\right)dx}{\int_0^L \frac{Kx^2}{L}dx} \equiv \frac{3L}{4}$$

19. **Ans (C)**

According to problem disc is melted and recasted into a solid sphere so their volume will be same

$$V_{Disc} = V_{Sphere} \Rightarrow \pi R^2_{Disc}\, t = \frac{4}{3}\pi R^3_{Sphere}$$

$$\Rightarrow \quad \pi R^2_{Disc}\left(\frac{R_{Disc}}{6}\right) = \pi R^3_{Sphere}$$

$\left[t = \frac{R_{Disc}}{6}, \text{given}\right]$

$$\Rightarrow \pi R^3_{Disc} = 8R^3_{Sphere} \Rightarrow R_{Sphere} = \frac{R_{Disc}}{2}$$

Moment of inertia of disc $I_{Disc} = \frac{1}{2} MR^2_{Disc} = I$.

Given $M(R_{Disc})^2 = 2I$

Moment of inertia of sphere

$$I_{sphere} = \frac{2}{5} NR^2_{Sphere}$$

$$= \frac{2}{5} M\left(\frac{R_{Disc}}{2}\right)^2 = \frac{M}{10}(R_{Disc})^2 = \frac{2I}{10} = \frac{I}{5}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (2)**

$$\gamma = \frac{\Delta V}{V\,\Delta T}$$

Given $\frac{\Delta V}{V} = 0.12\%$ or 0.0012

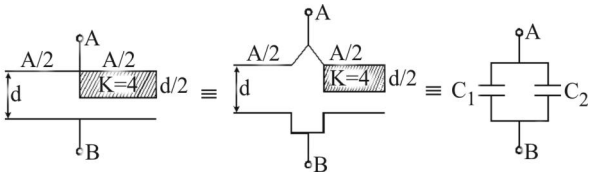
$$\gamma = \frac{0.0012}{20} = 6 \times 10^{-5}/^\circ C$$

Since, $\alpha = \frac{\gamma}{3} = 2 \times 10^{-5}/^\circ C$

3. **Ans (10)**

$$\phi = \vec{E} \cdot \vec{S} = \left(5\hat{i} + 2\hat{j}\right) \cdot 2\hat{i}$$

4. **Ans (23)**



$$C_1 = \frac{\varepsilon_0 A/2}{d}, \quad C_2 = \frac{\varepsilon_0 A/2}{\frac{d/2}{k} + \frac{d}{2}} = \frac{4\varepsilon_0 A}{5d}$$

$$C = C_1 + C_2 = \frac{13}{10}\frac{\varepsilon_0 A}{d}$$

Ans. $\frac{13}{10}\frac{\epsilon_0 A}{d}$

5. **Ans (6)**

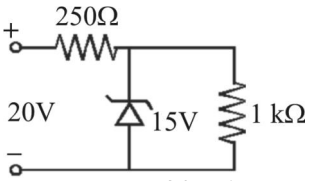
Binding energy of $_{11}N^{23}$ nucleus

$= 0.21 \times 931.5$ MeV

Binding energy per nucleon

$$= \frac{0.21 \times 931.5}{23} = 8.505 \text{ MeV}$$

6. **Ans (5)**



$$I_{Zener\ diode} = \frac{20-15}{250} - \frac{15}{1000}$$

$$= \frac{5}{250} - \frac{3}{200}$$

$$= \frac{1}{50} - \frac{3}{200} = \frac{4-3}{200}$$

$$= \frac{1}{200} = 5\,mA$$

7. **Ans (3)**

$$\frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} = \frac{1.5}{2} = \left(\frac{3}{4}\right)$$

$$\sin i_c = \frac{\mu_2}{\mu_1} = \frac{3}{4}$$

$$\tan i_c = \frac{3}{\sqrt{7}}$$

8. **Ans (396)**

$$n \propto \frac{1}{\ell} \Rightarrow n_1\ell_1 = n_2\ell_2 \Rightarrow (n+4)49 = (n-4)50$$

$$\Rightarrow n = 396$$

9. **Ans (2)**

The pendulums will be in phase if

$(N + 1)\, T_1 = NT_2$

N : no. of osc. completed by large pendulum

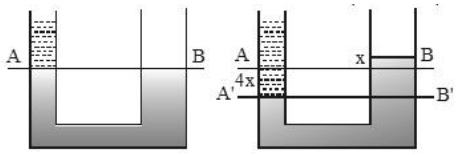
$(N + 1)\, T = N.\, 2T \Rightarrow N = 1$

no. of osc. by shorter pendulum = N + 1 = 2

10. **Ans (56)**

If the rise of level in the right limb be x cm. the fall of level of mercury in left limb be 4x cm because the area of cross section of right limb is 4 times as that of left limb.

∴ Level of water in left limb is (36 + 4x) cm.



Now equating pressure at interface of Hg and water (at A'B')

$(36 + 4x) \times 1 \times g = 5x \times 13.6 \times g$

By solving we get x = 0.56 cm

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (D)**

By decreasing volume of container $Q_C \uparrow > K_C$ then reaction moves in backward direction (α) decreases but K_P depends only on temp.

2. **Ans (D)**

Rate of reaction depends on stoichiometry of reaction. $N_2 + 3H_2 \longrightarrow 2\,NH_3$.

$$\frac{(ROD)_{H_2}}{3} = \frac{(ROA)_{NH_3}}{2}$$

$$(ROD)_{H_2} = \frac{(1.7 \times 1000)/17 \times 60}{2}$$

$$(ROD)_{H_2} = \frac{3}{2} \times \frac{1.7 \times 1000}{17 \times 60} \times 2 = 5 \text{ gm/min.}$$

3. **Ans (A)**

Total volume = 500 + 1500 = 2000 ml = 2L

Moles of $[Na^+]$ ions = $0.2 \times 0.5 = 0.1$

Concentration of $\left[Na^+\right]$ ions $= \frac{0.1}{2} = 0.05M$

Moles of $[Mg^{2+}]$ ions = $0.4 \times 1.5 = 0.6$

Concentration of $[Mg^{2+}]$ ions

$= \frac{0.6}{2} = 0.3M = 7.2gm/L$

Moles of $[Cl^-]$ ions = $0.2 \times 0.5 + 2 \times 0.4 \times 1.5 = 1.3$

Concentration of $[Cl^-]$ ions $= \frac{1.3}{2} = 0.65$ M

5. **Ans (C)**

$$\frac{P^o_{solvent} - P_s}{P^o_{solvent}} = \frac{w_{solute} \times (mol.\ wt)_{solvent}}{(mol\ wt)_{solute} \times w_{solvent}}$$

(for very dilute solution)

$$\frac{854 - 848.9}{854} = \frac{2 \times 76}{(mol.\ wt)_{solute} \times 100}$$

$(mol.\ wt)_{solute} = 254.5$

mol. wt of $S_n = 32 \times n$

∴ $32 \times n = 254.5$

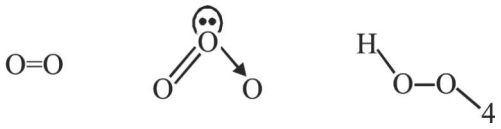
$n = 7.95$

$= 8$

8. **Ans (D)**

Nb belongs to d - block.

10. Ans (C)



B.O. = 2 B.O. = 1.5 B.O. = 1

$$B.O. \propto \frac{1}{B.L}$$

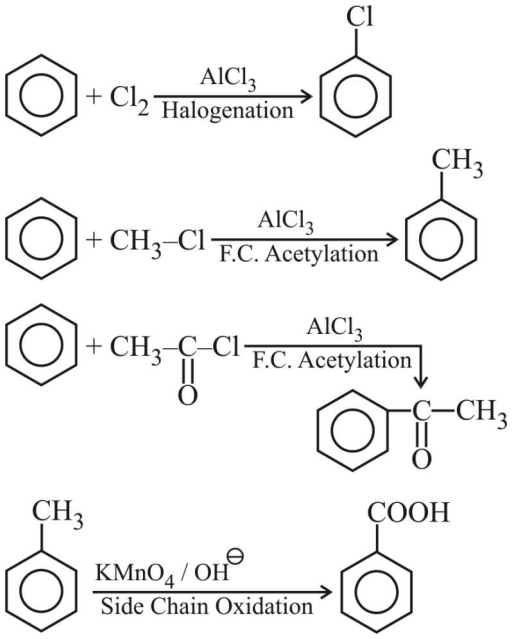
B.L. $\longrightarrow O_2 < O_3 < H_2O_2$

13. Ans (D)

For d orbital $\ell = 2$

$\Rightarrow \sqrt{\ell(\ell+1)}\hbar = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar$

15. Ans (A)



17. Ans (C)

P $\Rightarrow$ Cumene hydroperoxide to phenol.

Q $\Rightarrow$ Benzene diazonium chloride to benzene.

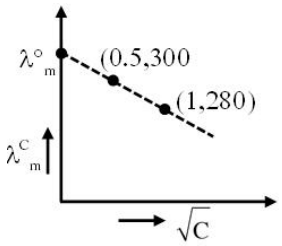
R $\Rightarrow$ Salicylic acid to benzoic acid.

S $\Rightarrow$ Paracetyl toluene to ptere phthalic acid.

PART-2 : CHEMISTRY

SECTION-II

1. Ans (320)



Equation of straight line

$$(y - 300) = \frac{280 - 300}{1 - 0.5}(x - 0.5)$$

$y - 300 = -40x + 20$

$y = -40x + 320$

2. Ans (3)

geq of $FeSO_4$ = geq of $K_2Cr_2O_7$

$\because$ mole × n factor = geq

$\therefore \quad 3 \times 1 = a \times 6 \quad ...(i)$

similarly geq of FeC_2O_4 = geq of $K_2Cr_2O_7$

$3 \times 3 = b \times 6 \quad ...(ii)$

$$\frac{3 \times 1}{3 \times 3} = \frac{a \times 6}{b \times 6}$$

$$\left[\frac{a}{b} = \frac{1}{3}\right]$$

3. Ans (48)

$$\text{moles of Al} = \frac{54}{27} = 2$$

$\because$ moles of Al = moles of Mg (given)

$\therefore$ moles of Mg = 2

mass of Mg = 2 × 24 = 48 gm

4. Ans (2)

$$R = 0.529 \times \frac{n^2}{Z}$$

$\Rightarrow Be^{3+} \Rightarrow Z = 4$

R = 0.529

n = 2

5. Ans (3)

Basic hydroxides –NaOH, KOH, $Ca(OH)_2$

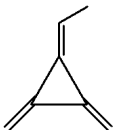
$Zn(OH)_2$, $Al(OH)_3$ $\Rightarrow$ Amphoteric

HOCl, H_2SO_4, HNO_3 $\Rightarrow$ Acidic

8. Ans (6)

i, iv, v, vi, vii, x

10. Ans (0)



can not show geometrical isomerism.

PART-3 : MATHEMATICS

SECTION-I

2. Ans (B)

$$\lim_{n\to\infty}\sum_{r=1}^{n}(\alpha^r+\beta^r)=\sum_{r=1}^{\infty}\alpha^r+\sum_{r=1}^{\infty}\beta^r$$

$$=\frac{\alpha}{1-\alpha}+\frac{\beta}{1-\beta}$$

$$=\frac{\alpha-\alpha\beta+\beta-\alpha\beta}{1-(\alpha+\beta)+\alpha\beta}$$

$$=\frac{\alpha+\beta-2\alpha\beta}{1-(\alpha+\beta)+\alpha\beta}$$

$$=\frac{\frac{25}{375}+\frac{4}{375}}{1-\frac{25}{375}-\frac{2}{375}}=\frac{29}{348}=\frac{1}{12}$$

3. Ans (B)

$|\text{adj adj adj adj adj }(2A)| = |2A|^{(3-1)^5}$

if $|A| = 2$ so $|2A| = 16$

$\Rightarrow (16)^{32} = (24)^{32} = 2^{128}$

4. Ans (D)

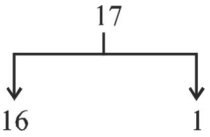
$(1-6)(2-7)(3-8)(4-9)$

$(5-10)(6-11)(7-12)(8-13)$

there are ${}^8C_1 \times 2 \times {}^7C_2 \times 2$ ways

$32 \times {}^7C_2$

5. Ans (C)



5 RedC.C, 11 other ; 1 Red C.C.

$$\frac{{}^6C_5\times{}^{46}C_{11}}{{}^{52}C_{16}}\times\frac{1}{{}^{36}C_1}$$

$$\frac{{}^{46}C_{11}}{6\times{}^{52}C_{16}}$$

6. Ans (D)

Let $z = i^{[1+3+5+...+(2n+1)]}$

Clearly series is A.P. with common difference = 2

$\because T_n = 2n - 1$ and $T_{n+1} = 2n+1$

So, number of terms in A.P. = (n + 1)

Now, $S_{n+1} = \frac{n+1}{2}[2.1 + (n+1-1)2]$

$\Rightarrow S_{n+1} = \frac{n+1}{2}[2+2n] = (n+1)^2$.

$\therefore \quad z = i^{(n+1)^2}$

Now put n = 1,2,3,4,5 .....

$n = 1, z = i^4 = 1$, $n = 2, z = i^9 = i$,

$n = 3, z = i^{16} = 1$, $n = 4, z = i^{25} = i$

7. Ans (B)

$y = 7\cos^2x + 6\sin x\cos x - \sin^2x$

$= 3\sin 2x + 4\cos 2x + 3$

The maximum value of y is 8

Also $\log_{\sqrt{2}}^{8} = \log_{\sqrt{2}}^{(\sqrt{2})^6} = 6$

8. Ans (D)

$\sigma = 2, \sum_{i=1}^{n} x_i = 20, \sum_{i=1}^{n} x_i^2 = 100$

$$\sigma = \sqrt{\frac{1}{n}\Sigma x_i^2 - \left(\frac{\Sigma x_i}{n}\right)^2}$$

$$(2)^2 = \frac{100}{n} - \left(\frac{20}{n}\right)^2$$

$$4 = \frac{100}{n} - \frac{400}{n^2}$$

$4n^2 = 100n - 400$

$= n^2 - 25n + 100 = 0$

$(n-5)(n-20) = 0$

$n = 5$ or 20

9. Ans (A)

$|a - a| = 0 < 1 \quad \therefore \; a\,R\,a \;\forall\; a \in R$

$\therefore$ R is reflexive

Again $a\,R\,b \quad \Rightarrow \quad |a-b| \le 1$

$\Rightarrow \quad |b-a| \le 1$

$\Rightarrow \quad b\,R\,a \quad \therefore$ R is symmetric

further 1 R 2 and 2 R 3 but $1 \not R 3$

$\therefore$ R is not transitive

10. Ans (B)

$f'(x) = \frac{-2}{\sqrt{1-x^2}} - \frac{4}{1+x^2} - 6x - 2$

$= -2\left[\frac{1}{\sqrt{1-x^2}} + \frac{2}{1+x^2} + 3x + 1\right]$

$f'(x) < 0 \Rightarrow f(x)$ is a dec. function

$f(1) = \pi + 5$

$f(-1) = 5\pi + 9$

Range : $[a, b] \equiv [\pi + 5, 5\pi + 9]$

$a = \pi + 5, b = 5\pi + 9 \Rightarrow 4a - b = 11 - \pi$

11. Ans (A)

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right) + \cos^{-1}\left(\cos\frac{7\pi}{6}\right) + \tan^{-1}\tan\left(\frac{3\pi}{4}\right)$

$\sin^{-1}\sin\left(\frac{2\pi}{3}\right) = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$

$\cos^{-1}\left(\cos\frac{2\pi}{6}\right) = 2\pi - \frac{7\pi}{6} = \frac{5\pi}{6}$

$\tan^{-1}\tan\left(\frac{3\pi}{4}\right) = \frac{3\pi}{4} - \pi = \frac{-\pi}{4}$

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right) + \cos^{-1}\left(\cos\frac{7\pi}{6}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

$= \frac{11\pi}{12}$

12. Ans (C)

$f(1) = 3$

For $x < 1$, $f'(x) = 3x^2 - 2x + 10 > 0$

$\Rightarrow f(x)$ is increasing

For $x > 1$, $f'(x) < 0$

function is decreasing.

$\lim\limits_{x\to 1^+} f(x) = -2 + \log_2(b^2 - 4)$

For maximum value at $x = 1$

$3 \geqslant -2 + \log_2(b^2 - 4)$

$32 \geqslant b^2 - 4 > 0$

$b \in [-6, -2) \cup (2, 6]$

13. Ans (D)

$f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$

$f_\lambda'(x) = 12\lambda x^2 - 72\lambda x + 36$

$f_\lambda'(x) = 12(\lambda x^2 - 6\lambda x + 3) \geq 0$

$\therefore \lambda > 0$ & $D \leq 0$

$36\lambda^2 - 4 \times \lambda \times 3 \leq 0$

$9\lambda^2 - 3\lambda \leq 0$

$3\lambda(3\lambda - 1) \leq 0$

$\lambda \in \left[0, \frac{1}{3}\right]$

$\therefore \lambda_{largest} = \frac{1}{3}$

$f(x) = \frac{4}{3}x^3 - 12x^2 + 36x + 48$

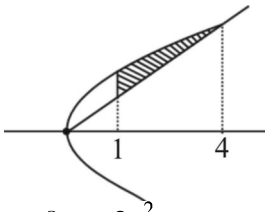
$\therefore f(1) + f(-1) = 72$

14. Ans (B)

$y^2 = 8x \quad \ldots(1)$

$y = \sqrt{2}x \quad \ldots(2)$

$y^2 = 2x^2$



$\Rightarrow 8x = 2x^2$

$\Rightarrow x = 0$ & 4

Area : $= \int_1^4 2\sqrt{2}\sqrt{x} - \sqrt{2}x\,dx$

$= 2\sqrt{2}\left(\frac{x^{\frac{3}{2}}}{3/2}\right)_1^4 - \sqrt{2}\left(\frac{x^2}{2}\right)_1^4$

$= \frac{4\sqrt{2}}{3}(8-1) - \frac{\sqrt{2}}{2}(16-1)$

$= \frac{28\sqrt{2}}{3} - \frac{15\sqrt{2}}{2} = \frac{11\sqrt{2}}{6}$

15. Ans (B)

$$xdy = \left(\sqrt{x^2+y^2}+y\right)dx$$

$$xdy - ydx = \sqrt{x^2+y^2}dx$$

$$\frac{xdy-ydx}{x^2} = \sqrt{1+\frac{y^2}{x^2}}\cdot\frac{dx}{x}$$

$$\frac{d(y/x)}{\sqrt{1+\left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$$

$$\ln\left(\frac{y}{x}+\sqrt{\left(\frac{y}{x}\right)^2+1}\right) = \ln x + R$$

$$\frac{y+\sqrt{y^2+x^2}}{x} = cx$$

$$y+\sqrt{y^2+x^2} = cx^2$$

$x = 1, y = 0 \Rightarrow 0 + 1 = C \Rightarrow C = 1$

Curve is $y+\sqrt{x^2+y^2} = x^2$

$x = 2, y = \alpha$

$$\alpha+\sqrt{4+\alpha^2} = 4$$

$$4+\alpha^2 = 16+\alpha^2-8\alpha$$

$$\alpha = \frac{3}{2}$$

16. Ans (A)

$$I(x) = \underset{II}{\int \sec^2 x.\sin^{-2022}x\,dx} - 2022\underset{I}{\int \sin^{-2022}x\,dx}$$

$$= \tan x.(\sin x)^{-2022} + \int (2022)\tan x.(\sin x)^{-2023}\cos x dx$$

$$I(x) = (\tan x)(\sin x)^{-2022} + C$$

$$\text{At } X = \pi/4,\ 2^{1011} = \left(\frac{1}{\sqrt{2}}\right)^{-2022} + C \ \therefore C = 0$$

$$\text{Hence } I(x) = \frac{\tan x}{(\sin x)^{2022}}$$

$$I(\pi/6) = \frac{1}{\sqrt{3}\left(\frac{1}{2}\right)^{2022}} = \frac{2^{2022}}{\sqrt{3}}$$

$$I(\pi/3) = \frac{\sqrt{3}}{\left(\frac{\sqrt{3}}{2}\right)^{2022}} = \frac{2^{2022}}{\left(\sqrt{3}\right)^{2021}} = \frac{1}{3^{1010}}I\left(\frac{\pi}{6}\right)$$

$$3^{1010}I(\pi/3) = I(\pi/6)$$

17. Ans (A)

$$|\vec{b}|^2 = 6 : |\vec{a}||\vec{b}|\cos\theta = 3\sqrt{2}$$

$$|\vec{a}|^2\ |\vec{b}|^2\cos^2\theta = 18$$

$$|\vec{a}|^2 = 6$$

Also $1+\alpha^2+\beta^2 = 6$

$$\alpha^2+\beta^2 = 5$$

to find

$$(\alpha^2+\beta^2)|\vec{a}|^2|\vec{b}|^2\sin^2\theta$$

$$= (5)(6)(6)\left(\frac{1}{2}\right) = 90$$

18. Ans (C)

$$e_h = \sqrt{1+\sin^2\theta}$$

$$e_c = \sqrt{1-\sin^2\theta}$$

$$e_h = \sqrt{7}e_c$$

$$1+\sin^2\theta = 7(1-\sin^2\theta)$$

$$\sin^2\theta = \frac{6}{8} = \frac{3}{4}$$

$$\sin\theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{3}$$

19. Ans (D)

$x^2 + y^2 = 4$

C (0, 0) $r_1 = 2$

C' $(2\lambda, 0)$ $r_2 = \sqrt{4\lambda^2 - 9}$

$|r_1 - r_2| < CC' < |r_1 + r_2|$

$\left|2 - \sqrt{4\lambda^2 - 9}\right| < |2\lambda| < 2 + \sqrt{4\lambda^2 - 9}$

$4 + 4\lambda^2 - 9 - 4\sqrt{4\lambda^2 - 9} < 4\lambda^2$

True $\lambda \in R$ .... (1)

$4\lambda^2 < 4 + 4\lambda^2 - 9 + 4\sqrt{4\lambda^2 - 9}$

$5 < 4\sqrt{4\lambda^2 - 9}$ and $\lambda^2 \geq \frac{9}{4}$

$\frac{25}{16} < 4l^2 - 9 \quad \lambda \in \left(-\infty, -\frac{3}{2}\right] \cup \left[\frac{3}{2}, \infty\right)$

$\frac{169}{64} < \lambda^2$

$\lambda \in \left(-\infty, -\frac{13}{8}\right) \cup \left(\frac{13}{8}, \infty\right)$ ...(2)

from (1) and (2)

$\lambda \in \left(-\infty, -\frac{13}{8}\right) \cup \left(\frac{13}{8}, \infty\right)$

$\Rightarrow R - \left[-\frac{13}{8}, \frac{13}{8}\right]$

as per question $a = -\frac{13}{8}$ and $b = \frac{13}{8}$

$\therefore$ required point is $(-1, 6)$ with satisfies option (4)

20. Ans (C)

Pt on $y^2 = 4\lambda$

$P(t^2, 2 + 1)$

centre y circle C(2, 8)

Let $z = d^2 = (t^2 - 2)^2 + (2t - 8)^2$

$\frac{dz}{dt} = 2(t^2 - 2).\ 2t + 2(2t - 8).\ 2 = 0$

$4[t^3 - 2t + 2t - 8] =$

$t^3 = 8 \Rightarrow t = 2$

P(4, 4)

$d^2 = (4 - 2)^2 + (4 - 8)^2$

$= 4 + 16 = 20$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1)

$\Delta = (2\cos x + \sin x)$

$$\begin{vmatrix} 1 & \cos x & \cos x \\ 1 & \sin x & \cos x \\ 1 & \cos x & \sin x \end{vmatrix} (C_1 \to C_1 + C_2 + C_3)$$

$$= (2\cos x + \sin x) \begin{vmatrix} 0 & \cos x - \sin x & 0 \\ 1 & \sin x & \cos x \\ 1 & \cos x & \sin x \end{vmatrix} (R_1 \to R_1 - R_2)$$

$= (2\cos x + \sin x)(\cos x - \sin x)2 = 0$

$\therefore \tan x = -2$ or 1; Hence one solution

2. Ans (0)

$T_{r+1} = {}^{65}C_r 5^{(65-r)\frac{1}{2}} 7^{r/6}$

$\therefore \quad \frac{65 - r}{2} \in I \quad \& \quad \frac{r}{6} \in I$

r = 1, 3, 5, 7, ..., 65

& also, r = 0, 6, 12, 18, 24, ...60

No. common 'r'

$\therefore$ total integral terms = 0

3. Ans (3)

$\left(1 + \sqrt{3} + \tan 1^\circ\right)\left(1 + \sqrt{3} + \tan 2^\circ\right)$

$(\tan 1^\circ + \tan(60^\circ - 1^\circ))$

$\Rightarrow \frac{(\tan 1^\circ + \tan(60^\circ - 2^\circ))}{\left(1 + \tan^2 1^\circ\right)\left(1 + \tan^2 2^\circ\right)}$

$= (1 + \sqrt{3}\tan 1^\circ)(1 + \sqrt{3}\tan 2^\circ)$

$$\frac{\left(\tan 1^\circ + \frac{\sqrt{3} - \tan 1^\circ}{1 + \sqrt{3}\tan 1^\circ}\right)\left(\tan 2^\circ + \frac{\sqrt{3} - \tan 2^\circ}{1 + \sqrt{3}\tan 2^\circ}\right)}{(1 + \tan^2 1^\circ)(1 + \tan^2 2^\circ)}$$

$$(1 + \sqrt{3}\tan 1^\circ)(1 + \sqrt{3}\tan 2^\circ)$$

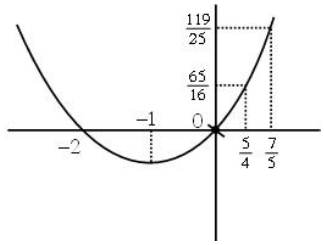
$$= \frac{(\sqrt{3}\tan^2 1^\circ + \sqrt{3})(\sqrt{3}\tan^2 2^\circ + \sqrt{3})}{(1 + \tan^2 1^\circ)(1 + \tan^2 2^\circ)}$$

$$\left(1 + \sqrt{3}\tan 1^\circ\right)\left(1 + \sqrt{3}\tan 2^\circ\right)$$

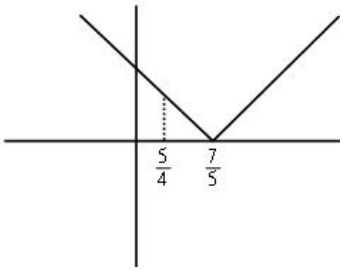
$\sqrt{3} \cdot \sqrt{3} = 3$

4. Ans (15)

Graph of $x^2 + 2x$



Graph of $|5x - 7|$



$f(x)|_{min} = 4 + 0 = 4$, at $x = \frac{7}{5}$

$f(x)|_{max} = 8 + 3$, at $x = 2$

Required sum = 15

5. Ans (3)

$$f(x) = \begin{cases} (x^2-1)(x-3)+(x-3),\ x \in (0,1] \cup [3,4) \\ -(x^2-1)(x-3)+(x-3),\ x \in [1,3] \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} 3x^2-6x,\ x \in (0,1) \cup (3,4) \\ -3x^2+6x+2,\ x \in (1,3) \end{cases}$$

f(x) is non-derivable at x = 1 and x = 3(minima)

also f'(x) = 0 at $x = 1 + \sqrt{\frac{5}{3}}$ m + M = 3 (max.)

6. Ans (6)

$$I = \frac{48}{\pi^4}\int_0^{\pi} x^2\left(\frac{3\pi}{2}-x\right)\frac{\sin x}{1+\cos^2 x}\,dx \quad ...(1)$$

Apply king property

$$I = \frac{48}{\pi^4}\int_0^{\pi} (\pi-x)^2\left(\frac{\pi}{2}+x\right)\frac{\sin x}{1+\cos^2 x}\,dx \quad ...(2)$$

(1) + (2)

$$I = \frac{12}{\pi}\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}\,dx$$

Let cosx = t $\Rightarrow$ sinxdx = −dt

$$I = \frac{12}{\pi}\int_1^{-1} \frac{-dt}{1+t^2} = 6$$

9. Ans (22)

$$\frac{x-4}{12} = \frac{x+6}{4} = \frac{z+2}{6}$$

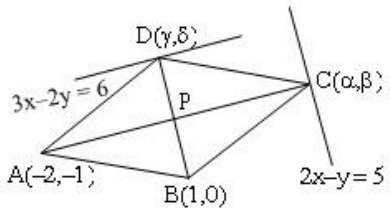
$$\frac{x-4}{\frac{6}{7}} = \frac{y+6}{\frac{2}{7}} = \frac{z+2}{\frac{3}{7}} = 21$$

$$\left(21\times\frac{6}{7}+4, \frac{2}{7}\times 21-6, \frac{3}{7}\times 21-2\right)$$

= (22, 0, 7) = (a, b, c)

$\therefore \sqrt{324+144+16} = 22$

10. Ans (32)



$$P \equiv \left(\frac{\alpha-2}{2}, \frac{\beta-1}{2}\right) \equiv \left(\frac{\gamma+1}{2}, \frac{\delta}{2}\right)$$

$$\frac{\alpha-2}{2} = \frac{\gamma+1}{2} \text{ and } \frac{\beta-1}{2} = \frac{\delta}{2}$$

$\Rightarrow \alpha-\gamma = 3$.....(1), $\beta-\delta = 1$.....(2)

Also, (γ, δ) lies on $3x - 2y = 6$

$3\gamma - 2\delta = 6$ . . . . . (3)

and (α, β) lies on $2x - y = 5$

$\Rightarrow 2\alpha-\beta = 5$.....(4)

Solving (1), (2), (3), (4)

$\alpha = -3,\ \beta = -11, \gamma = -6, \delta = -12$

$|\alpha+\beta+\gamma+\delta| = 32$